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Deep Generative Models

Lecture 10

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Recap of Previous Lecture

Continuous-Time Dynamics

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0.$$

$$\boldsymbol{\psi}_t(\mathbf{x}_0) = \int_{t_0}^{t} \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(s), s)ds + \mathbf{x}_0 \approx \texttt{ODESolve}_v(\mathbf{x}_0, \boldsymbol{\theta}, t_0, t).$$

- $\mathbf{v}_{\boldsymbol{\theta}} : \mathbb{R}^m \times [t_0, t_1] \to \mathbb{R}^m$ is a vector field.
- $\boldsymbol{\psi} : \mathbb{R}^m \times [t_0, t_1] \to \mathbb{R}^m$ is a flow (the solution of ODE):

Euler `ODESolve`

$$\mathbf{x}(t + h) = \mathbf{x}(t) + h \cdot \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)$$

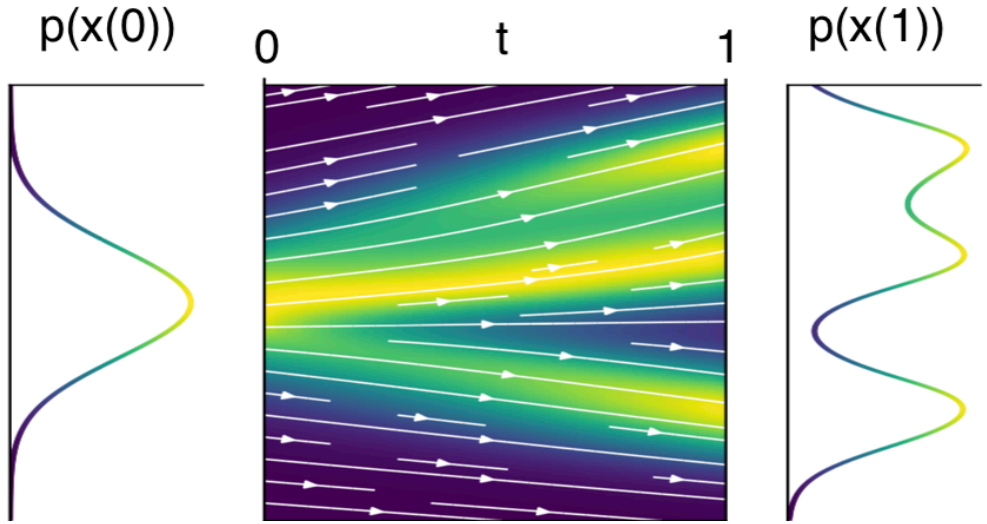
More advanced numerical methods (such as Runge-Kutta) are often used instead of unstable Euler update step.

Chen R. T. Q. et al. Neural Ordinary Differential Equations, 2018

Recap of Previous Lecture

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t); \quad \mathbf{x}(t_0) = \mathbf{x}_0$$

- Suppose $\mathbf{x}(0)$ is a random variable with density $p_0(\mathbf{x})$. Then, $\mathbf{x}(t)$ is a random variable with density $p_t(\mathbf{x})$.
- $p_t(\mathbf{x}) = p(\mathbf{x}, t)$ describes the **probability path** between $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$.



Recap of Previous Lecture

Theorem (Picard)

If $\mathbf{v}$ is continuously differentiable with a bounded derivative in $\mathbf{x}$ and continuous in t, then the ODE has a **unique solution** given by a flow $\boldsymbol{\psi}_t$.

This guarantees the ODE is **uniquely reversible**.

$$\boldsymbol{\psi}_1(\mathbf{x}_0) = \mathbf{x}_0 + \int_0^1 \mathbf{v}_{\boldsymbol{\theta}}(\boldsymbol{\psi}_t(\mathbf{x}_0), t)dt$$

$$\mathbf{x}(1) = \mathbf{x}(0) + \int_0^1 \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)dt$$

$$\mathbf{x}(0) = \mathbf{x}(1) + \int_1^0 \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)dt$$

Note: Unlike discrete-time flows, $\mathbf{v}$ need not be invertible (uniqueness ensures bijection).

How can we compute $p_t(\mathbf{x})$ at arbitrary t?

Recap of Previous Lecture

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0$$

Theorem (Continuity Equation)

If $\mathbf{v}$ is continuously differentiable in $\mathbf{x}$ (with bounded derivative) and continuous in t, and $p_t(\mathbf{x}) > 0$, then

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

Solution

$$\log p_1(\mathbf{x}(1)) = \log p_0(\mathbf{x}(0)) - \int_0^1 \operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right) dt.$$

- This solution gives us the density **along the trajectory**.
- However, it's difficult to efficiently estimate the last term.

Recap of Previous Lecture

SDE Basics

Let's define a stochastic process $\mathbf{x}(t)$ with initial condition $\mathbf{x}(0) \sim p_0(\mathbf{x})$:

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w},$$

where $\mathbf{w}(t)$ is the standard Wiener process (Brownian motion):

$$\mathbf{w}(t) - \mathbf{w}(s) \sim \mathcal{N}(0, (t-s)\mathbf{I}), \quad d\mathbf{w} = \boldsymbol{\epsilon} \cdot \sqrt{dt}, \text{ where } \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}).$$

Discretizing the SDE (Euler-Maruyama Update) – `SDESolve`

$$\mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{f}(\mathbf{x}(t), t) \cdot dt + g(t) \cdot \boldsymbol{\epsilon} \cdot \sqrt{dt}$$

- At each time t, we have the density $p_t(\mathbf{x}) = p(\mathbf{x}, t)$.
- $p : \mathbb{R}^m \times [0, 1] \to \mathbb{R}_+$ is a **probability path** between $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$.

Recap of Previous Lecture

Theorem (Kolmogorov-Fokker-Planck)

If $p_t(\mathbf{x}) \in C^{1,2}$ (i.e., C^1 in t and C^2 in $\mathbf{x}$), then

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})\right) + \frac{1}{2}g^2(t)\Delta_{\mathbf{x}} p_t(\mathbf{x})$$

Langevin SDE (Special Case)

$$d\mathbf{x} = \frac{1}{2}\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})dt + 1 \cdot d\mathbf{w}$$

The density $p_t(\mathbf{x})$ is a **stationary** distribution for the SDE. Setting the stationary density to the model distribution, $p_t(\mathbf{x}) = p_{\boldsymbol{\theta}}(\mathbf{x})$ for all t:

Langevin Dynamic

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}, \quad \eta \approx dt.$$

Outline

01 Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)

02 Probability Flow ODE

03 Reverse SDE

04 Score-Based Generative Models Through SDEs

05 Denoiser Parameterization and Sampling

06 Flow Matching (FM)

Outline

01 **Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)**

02 Probability Flow ODE

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Score Matching SDE

Denoising Score Matching

$$\mathbf{x}_t = \mathbf{x} + \sigma_t \cdot \boldsymbol{\epsilon}_t, \qquad q(\mathbf{x}_t|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_t^2 \cdot \mathbf{I})$$
$$\mathbf{x}_{t-1} = \mathbf{x} + \sigma_{t-1} \cdot \boldsymbol{\epsilon}_{t-1}, \quad q(\mathbf{x}_{t-1}|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_{t-1}^2 \cdot \mathbf{I})$$

$$\mathbf{x}_t = \mathbf{x}_{t-1} + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \cdot \boldsymbol{\epsilon}, \quad q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_{t-1}, (\sigma_t^2 - \sigma_{t-1}^2) \cdot \mathbf{I})$$

Let's transform this Markov chain into the continuous stochastic process $\mathbf{x}(t)$ by letting $T \to \infty$:

$$\begin{aligned}
\mathbf{x}(t) &= \mathbf{x}(t - dt) + \sqrt{\sigma^2(t) - \sigma^2(t - dt)} \cdot \boldsymbol{\epsilon} \\
&= \mathbf{x}(t - dt) + \sqrt{\frac{\sigma^2(t) - \sigma^2(t - dt)}{dt} dt} \cdot \boldsymbol{\epsilon} \\
&= \mathbf{x}(t - dt) + \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}
\end{aligned}$$

Score Matching SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

$$d\mathbf{x} = \mathbf{x}(t) - \mathbf{x}(t - dt)$$

Variance Exploding SDE

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

$\sigma(t)$ is a monotonically increasing function.

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = 0, \quad g(t) = \sqrt{\frac{d[\sigma^2(t)]}{dt}}$$

Diffusion SDE

Denoising Diffusion

$$\mathbf{x}_t = \sqrt{1-\beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \boldsymbol{\epsilon}_t, \quad q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I})$$

Let's turn this Markov chain into a continuous stochastic process by letting $T \to \infty$ and setting $\beta_t = \beta(\frac{t}{T}) \cdot \frac{1}{T}$ (where $dt = \frac{1}{T}$):

$$\begin{aligned}
\mathbf{x}(t) &= \sqrt{1-\beta(t)dt} \cdot \mathbf{x}(t-dt) + \sqrt{\beta(t)dt} \cdot \boldsymbol{\epsilon} \\
&\approx \left(1 - \frac{1}{2}\beta(t)dt\right) \cdot \mathbf{x}(t-dt) + \sqrt{\beta(t)dt} \cdot \boldsymbol{\epsilon} \\
&= \mathbf{x}(t-dt) - \frac{1}{2}\beta(t)\mathbf{x}(t-dt)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}
\end{aligned}$$

Diffusion SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) - \frac{1}{2}\beta(t)\mathbf{x}(t - dt)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

Variance Preserving SDE

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}(t), \quad g(t) = \sqrt{\beta(t)}$$

Variance is preserved as long as $\mathbf{x}(0)$ has unit variance.

Diffusion SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

Variance Exploding SDE (NCSN)

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

Variance Preserving SDE (DDPM)

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

We treat the discrete-time methods (NCSN, DDPM) as a special case of the continuous-time methods (VE-SDE, VP-SDE).

Lu C. et al. Dpm-solver: A fast ode solver for diffusion probabilistic model sampling in around 10 steps, 2022

Outline

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02 Probability Flow ODE

03 Reverse SDE

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05 Denoiser Parameterization and Sampling

06 Flow Matching (FM)

Probability Flow ODE

ODE and Continuity Equation

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt$$

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right) \Leftrightarrow \frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x}))$$

The only source of randomness is the initial distribution $p_0(\mathbf{x})$.

SDE and KFP Equation

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})) + \frac{1}{2}g^2(t)\Delta_{\mathbf{x}} p_t(\mathbf{x})$$

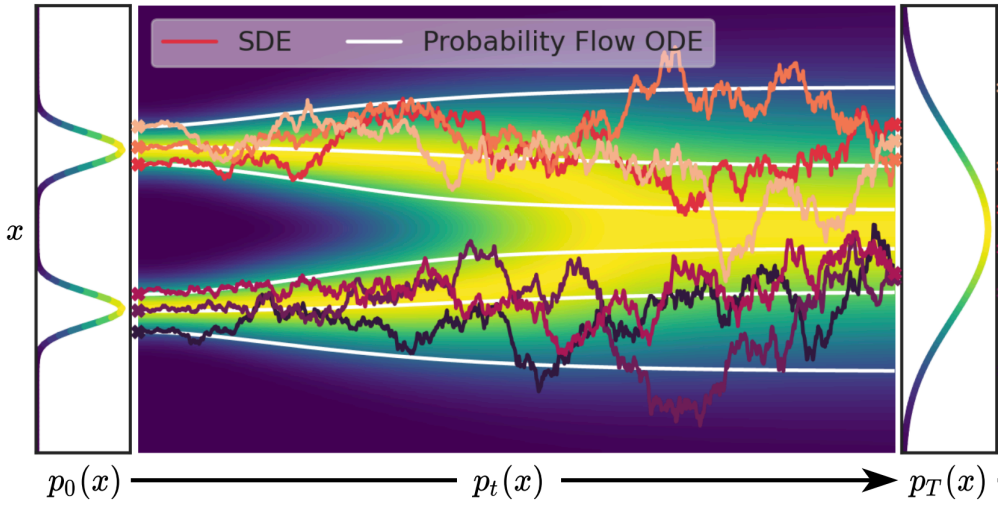
Now there are two sources of randomness: the initial distribution $p_0(\mathbf{x})$ and the Wiener process $\mathbf{w}(t)$.

Probability Flow ODE

Theorem

Suppose the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ induces the probability path $p_t(\mathbf{x})$ with $p_t(\mathbf{x}) > 0$. Then, there exists an ODE with the same probability path $p_t(\mathbf{x})$, given by

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right)dt$$



Probability Flow ODE

Theorem

Suppose the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ induces the probability path $p_t(\mathbf{x})$ with $p_t(\mathbf{x}) > 0$. Then, there exists an ODE with the same probability path $p_t(\mathbf{x})$, given by

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right)dt$$

Proof

$$\begin{aligned}
\frac{\partial p_t(\mathbf{x})}{\partial t} &= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})\right] + \frac{1}{2}g^2(t)\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2}\right) \\
&= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)\frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}}\right]\right) \\
&= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)p_t(\mathbf{x})\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right]\right) \\
&= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right)p_t(\mathbf{x})\right]\right)
\end{aligned}$$

Probability Flow ODE

Proof (Continued)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right)p_t(\mathbf{x})\right]\right)$$

$$= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x})\right]\right) = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x})\right)$$

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x}); \quad \tilde{g}(t) = 0$$

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt + 0 \cdot d\mathbf{w} = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right)dt$$

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x})) + \frac{1}{2}\tilde{g}^2(t)\Delta_{\mathbf{x}}p_t(\mathbf{x})$$

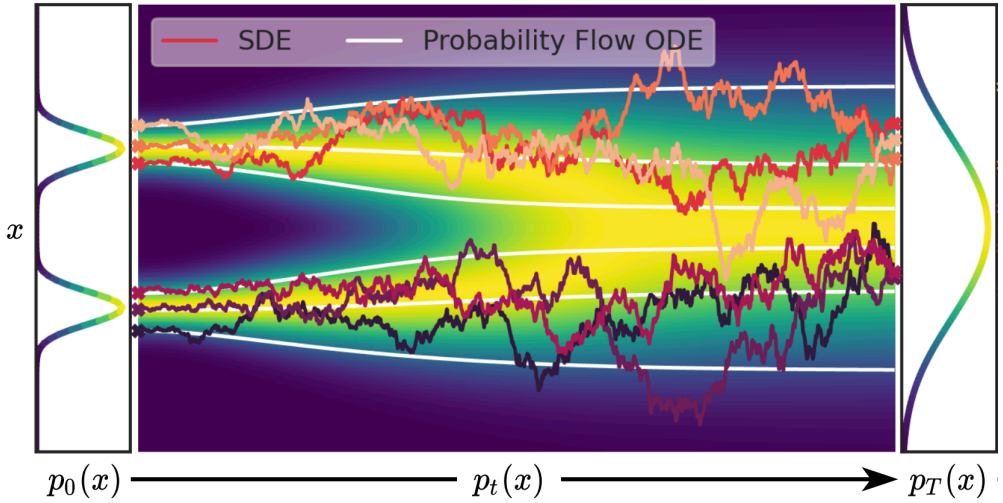
Probability Flow ODE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w} \quad - \text{SDE}$$

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt \quad - \text{Probability Flow ODE}$$

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$$

- $\mathbf{s}(\mathbf{x}, t) = \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$ is the continuous-time score function.
- The ODE produces more stable trajectories.



Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Why convert SDEs to PF-ODEs? (1) Efficient Sampling

Use the familiar ODE solvers with the learned PF-ODE field:

$$\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g^2(t) \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, t).$$

Solver	How it uses the network
Euler / Heun	Apply the familiar ODE updates: 1 / 2 evaluations per step. EDM uses Heun.
DPM-Solver **DPM-Solver++**	For VE/VP, integrate the linear drift exactly; approximate the network term. The ++ variant targets guided sampling.
UniPC	Correct the predicted ODE step using stored network predictions.

Compare **NFE** (network evaluations). DPM-Solver reports **10–20 NFE**; the required budget depends on the model.

Noise injection is a separate choice: it can improve quality, but does not help every model.

Karras T., et al. Elucidating the Design Space of Diffusion-Based Generative Models, 2022
Lu C., et al. DPM-Solver: A Fast ODE Solver for Diffusion Probabilistic Model Sampling in Around 10 Steps, 2022
Lu C., et al. DPM-Solver++: Fast Solver for Guided Sampling of Diffusion Probabilistic Models, 2022
Zhao W., et al. UniPC: A Unified Predictor-Corrector Framework for Fast Sampling of Diffusion Models, 2023

Why convert SDEs to PF-ODEs? (2) Deterministic Sampling

- The PF-ODE defines a deterministic, invertible map $\mathbf{x}(0) \leftrightarrow \mathbf{x}(T)$.
- Each terminal $\mathbf{x}(T)$ produces a unique trajectory back to data space.

Log-likelihood via the probability-flow ODE

$$\log p_0(\mathbf{x}(0)) = \log p_T(\mathbf{x}(T)) + \int_0^T \operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right) dt$$

Exact change-of-variables identity; numerical evaluation uses an ODE solver and, optionally, Hutchinson's trace estimator.

- Connects PF-ODE to continuous normalizing flows.
- Invertibility enables image $\rightarrow$ latent $\rightarrow$ edited image workflows.
- Smooth interpolation in latent space between two samples.

Chen R. T. Q., et al. Neural Ordinary Differential Equations, 2018
Grathwohl W. et al. FFJORD: Free-form Continuous Dynamics for Scalable Reversible Generative Models, 2018

Sampling: Reverse SDE vs PF-ODE

The same trained score $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, t)$ defines two sampling dynamics (reverse SDE derived next).

Sampling (Reverse SDE, Euler–Maruyama)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Initialize step $\Delta t = T/K$.
3. Denoise:

$$\mathbf{x}_{t-\Delta t} = \mathbf{x}_t - \Big[\mathbf{f}(\mathbf{x}_t, t) - g^2(t)\, \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)\Big]\Delta t + g(t)\sqrt{\Delta t}\cdot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$$

Sampling (PF-ODE, Euler)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Initialize step $\Delta t = T/K$.
3. Denoise:

$$\mathbf{x}_{t-\Delta t} = \mathbf{x}_t - \Big[\mathbf{f}(\mathbf{x}_t, t) - \tfrac{1}{2}\, g^2(t)\, \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)\Big]\Delta t \quad \text{(no noise injection)}$$

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Reverse SDE

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt, \qquad \mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{v}(\mathbf{x}, t)dt$$

Here dt can be > 0 or < 0.

Reverse ODE

Let $\tau = 1 - t$ ($d\tau = -dt$).

$$d\mathbf{x} = -\mathbf{v}(\mathbf{x}, 1 - \tau)d\tau$$

- How do we reverse the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$?
- The Wiener process introduces randomness that must be reversed.

Theorem

There exists a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ (assume $p_t(\mathbf{x}) > 0$), given by:

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})\right) dt + g(t)d\bar{\mathbf{w}}$$

where $dt < 0$ and $\bar{\mathbf{w}}$ is a reverse-time Wiener process.

Anderson B. D. O. Reverse-time diffusion equation models, 1982
Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Reverse SDE

Theorem

There exists a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ (assume $p_t(\mathbf{x}) > 0$), given by:

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})\right) dt + g(t)d\bar{\mathbf{w}}$$

where $dt < 0$ and $\bar{\mathbf{w}}(t)$ is a reverse-time Wiener process.

- Here the score function appears: $\mathbf{s}(\mathbf{x}, t) = \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$.
- Since $\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t)$ (PF-ODE), the reverse drift equals $\mathbf{f} - g^2\mathbf{s} = \mathbf{v} - \frac{1}{2}g^2\mathbf{s}$.

Proof Sketch

- Convert the initial SDE to a probability flow ODE.
- Reverse the probability flow ODE.
- Convert the reversed probability flow ODE back to an SDE.

Anderson B. D. O. Reverse-time diffusion equation models, 1982
Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Proof of Reverse SDE

- Convert the initial SDE to a probability flow ODE:

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right)dt$$

- Reverse the probability flow ODE:

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right)dt$$

$$d\mathbf{x} = \left(-\mathbf{f}(\mathbf{x}, 1-\tau) + \frac{1}{2}g^2(1-\tau)\frac{\partial}{\partial \mathbf{x}}\log p_{1-\tau}(\mathbf{x})\right)d\tau$$

Proof of Reverse SDE

- Convert the reversed probability flow ODE back to an SDE:

$$d\mathbf{x} = \left(-\mathbf{f}(\mathbf{x}, 1-\tau) + \frac{1}{2} g^2(1-\tau) \frac{\partial}{\partial \mathbf{x}} \log p_{1-\tau}(\mathbf{x}) \right) d\tau$$

$$d\mathbf{x} = \left(-\mathbf{f}(\mathbf{x}, 1-\tau) + g^2(1-\tau) \frac{\partial}{\partial \mathbf{x}} \log p_{1-\tau}(\mathbf{x}) \right) d\tau + g(1-\tau) d\mathbf{w}$$

Here $\mathbf{w}(\tau)$ is a Wiener process in forward τ-time.

Reverse SDE

Theorem

There exists a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ (assume $p_t(\mathbf{x}) > 0$), given by:

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})\right) dt + g(t)d\bar{\mathbf{w}}$$

where $dt < 0$ and $\bar{\mathbf{w}}(t)$ is a reverse-time Wiener process.

Proof (Continued)

$$d\mathbf{x} = \left(-\mathbf{f}(\mathbf{x}, 1-\tau) + g^2(1-\tau)\frac{\partial}{\partial \mathbf{x}} \log p_{1-\tau}(\mathbf{x})\right) d\tau + g(1-\tau)d\mathbf{w}$$

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})\right) dt + g(t)d\bar{\mathbf{w}}$$

Here $d\tau > 0$, $dt < 0$, and $\bar{\mathbf{w}}$ is reverse-time Wiener process.

Anderson B. D. O. Reverse-time diffusion equation models, 1982
Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

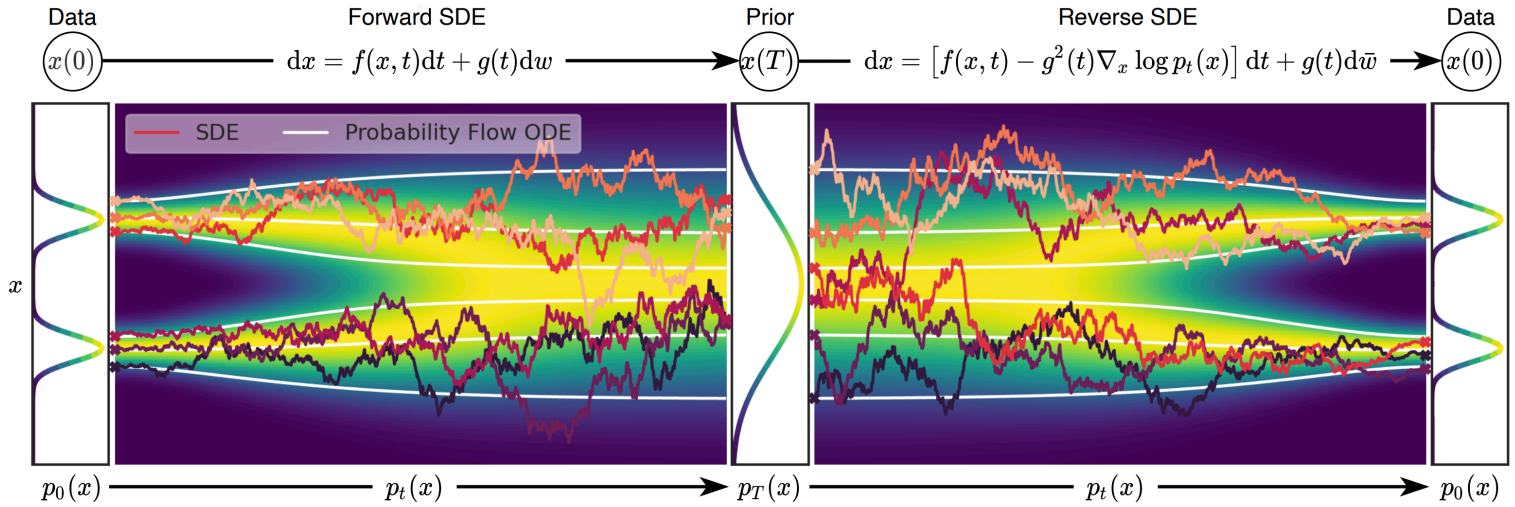
Reverse SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t), \quad \text{where} \quad \mathbf{s}(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$$

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt \quad \textbf{– Probability Flow ODE}$$

$$d\mathbf{x} = \left(\mathbf{v}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t)\right)dt + g(t)d\bar{\mathbf{w}} \quad \textbf{– Reverse SDE}$$



VE and VP: Probability Flow and Reverse SDE

Write $\mathbf{s} = \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$.

VE / NCSN

$\sigma(t)$ is monotonically increasing.

$$d\mathbf{x} = \sqrt{\frac{d\sigma^2(t)}{dt}}\, d\mathbf{w}$$

Probability Flow ODE

$$d\mathbf{x} = -\frac{1}{2}\frac{d\sigma^2(t)}{dt}\mathbf{s}\, dt$$

Reverse SDE

$$d\mathbf{x} = -\frac{d\sigma^2(t)}{dt}\mathbf{s}\, dt + \sqrt{\frac{d\sigma^2(t)}{dt}}\, d\bar{\mathbf{w}}$$

Here $dt < 0$ and $\bar{\mathbf{w}}$ is a reverse-time Wiener process.

VP / DDPM

Unit initial variance is preserved.

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}\, dt + \sqrt{\beta(t)}\, d\mathbf{w}$$

$$d\mathbf{x} = \left(-\frac{1}{2}\beta(t)\mathbf{x} - \frac{1}{2}\beta(t)\mathbf{s}\right) dt$$

$$d\mathbf{x} = \left(-\frac{1}{2}\beta(t)\mathbf{x} - \beta(t)\mathbf{s}\right) dt + \sqrt{\beta(t)}\, d\bar{\mathbf{w}}$$

Outline

01 Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)

02 Probability Flow ODE

03 Reverse SDE

04 Score-Based Generative Models Through SDEs

05 Denoiser Parameterization and Sampling

06 Flow Matching (FM)

Score-Based Generative Models Through SDEs

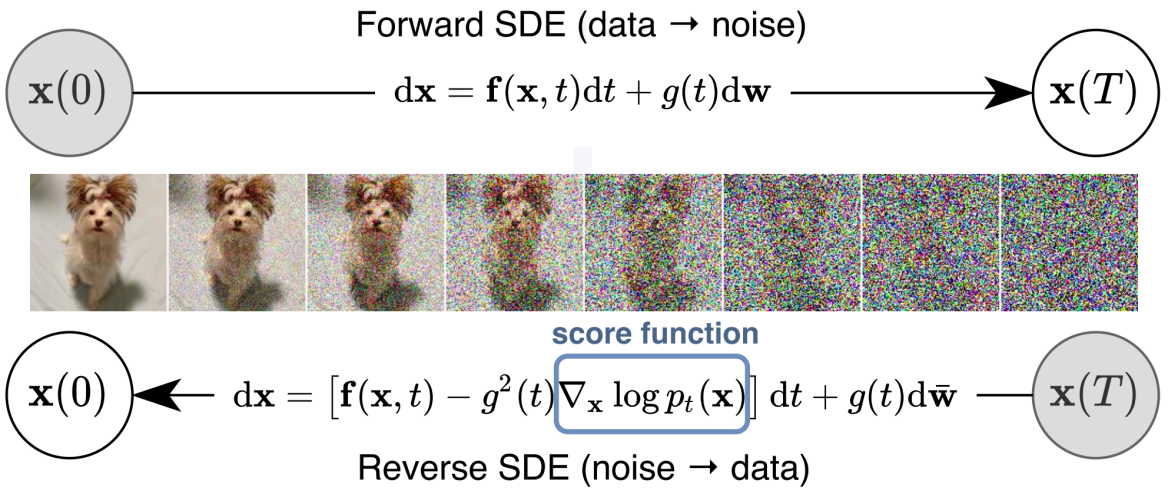
Discrete-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)}\mathbb{E}_{t\sim U\{1,T\}}\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left\|\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t}\log q(\mathbf{x}_t|\mathbf{x}_0)\right\|_2^2$$

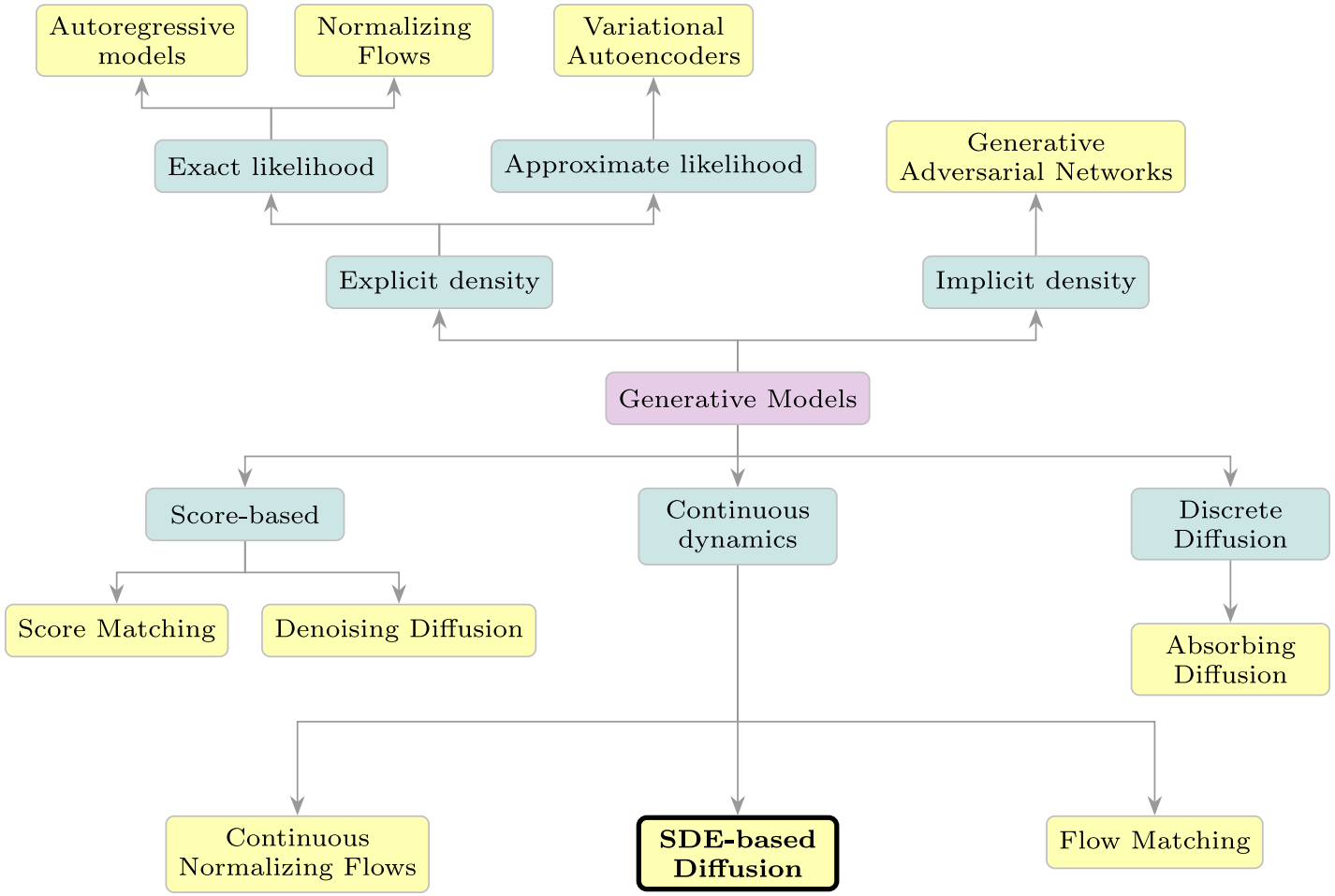
Is it possible to train score-based diffusion models in continuous time?

Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))}\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))}\left\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t),t) - \nabla_{\mathbf{x}(t)}\log q(\mathbf{x}(t)|\mathbf{x}(0))\right\|_2^2$$



Generative Models Taxonomy



Score-Based Generative Models Through SDEs

Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))}\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))}\left\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t),t) - \nabla_{\mathbf{x}(t)}\log q(\mathbf{x}(t)|\mathbf{x}(0))\right\|_2^2$$

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\Big(\boldsymbol{\mu}(\mathbf{x}(0),t), \sigma^2(t)\cdot\mathbf{I}\Big)$$

$$\nabla_{\mathbf{x}(t)}\log q(\mathbf{x}(t)|\mathbf{x}(0)) = -\frac{1}{\sigma^2(t)}(\mathbf{x}(t)-\boldsymbol{\mu})$$

Note: Normality holds for $\mathbf{f}(\mathbf{x},t)$ affine in $\mathbf{x}$ ($\mathbf{f}(\mathbf{x},t) = f(t)\mathbf{x}$).

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}}\cdot d\mathbf{w} \;\; \text{(Variance Exploding SDE)}$$

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)}\cdot d\mathbf{w} \;\; \text{(Variance Preserving SDE)}$$

Is it possible to explicitly derive $\boldsymbol{\mu}(\mathbf{x}(0),t)$ and $\sigma^2(t)$ for VE-SDE and VP-SDE?

Score-Based Generative Models Through SDEs

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\Big(\boldsymbol{\mu}(\mathbf{x}(0), t), \sigma^2(t) \cdot \mathbf{I}\Big)$$

Theorem

The moments of the SDE $d\mathbf{x} = f(t)\mathbf{x}dt + g(t)d\mathbf{w}$ satisfy:

$$\frac{d\boldsymbol{\mu}(\mathbf{x}(0), t)}{dt} = f(t)\boldsymbol{\mu}(\mathbf{x}(0), t), \qquad \frac{d\sigma^2(t)}{dt} = 2f(t)\sigma^2(t) + g^2(t)$$

Proof

$$\begin{aligned}
\mathbb{E}\left[d\mathbf{x}|\mathbf{x}(0)\right] &= \mathbb{E}\left[f(t)\mathbf{x}dt|\mathbf{x}(0)\right] + \mathbb{E}\left[g(t)d\mathbf{w}|\mathbf{x}(0)\right] \\
&= f(t)\mathbb{E}\left[\mathbf{x}|\mathbf{x}(0)\right]dt + g(t)\mathbb{E}\left[d\mathbf{w}|\mathbf{x}(0)\right] \\
&= f(t)\boldsymbol{\mu}(\mathbf{x}(0), t)dt \\
\frac{d\mathbb{E}\left[\mathbf{x}(t)|\mathbf{x}(0)\right]}{dt} &= \frac{d\boldsymbol{\mu}(\mathbf{x}(0), t)}{dt} = f(t)\boldsymbol{\mu}(\mathbf{x}(0), t)
\end{aligned}$$

We prove the mean equation only; the variance equation is stated without proof.

Score-Based Generative Models Through SDEs

Theorem

$$\frac{d\boldsymbol{\mu}(\mathbf{x}(0), t)}{dt} = f(t)\boldsymbol{\mu}(\mathbf{x}(0), t)$$

Examples

$$\textbf{NCSN:} \quad f(t) = 0 \quad \Rightarrow \quad \boldsymbol{\mu} = \mathbf{x}(0)$$

$$\textbf{DDPM:} \quad f(t) = -\frac{1}{2}\beta(t) \quad \Rightarrow \quad \frac{d\boldsymbol{\mu}}{dt} = -\frac{1}{2}\beta(t)\boldsymbol{\mu}$$

$$\boldsymbol{\mu} = \mathbf{x}(0)\exp\left(-\frac{1}{2}\int_0^t \beta(s)ds\right)$$

Score-Based Generative Models Through SDEs

Training

1. Sample $\mathbf{x}(0) \sim p_{\text{data}}(\mathbf{x})$, $t \sim U[0, 1]$.
2. Sample $\mathbf{x}(t) \sim q(\mathbf{x}(t)|\mathbf{x}(0))$.
3. Compute loss $\mathcal{L} = \left\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0))\right\|_2^2$.

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\Big(\boldsymbol{\mu}(\mathbf{x}(0), t), \sigma^2(t) \cdot \mathbf{I}\Big)$$

NCSN

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0), \left[\sigma^2(t) - \sigma^2(0)\right] \cdot \mathbf{I}\right)$$

DDPM

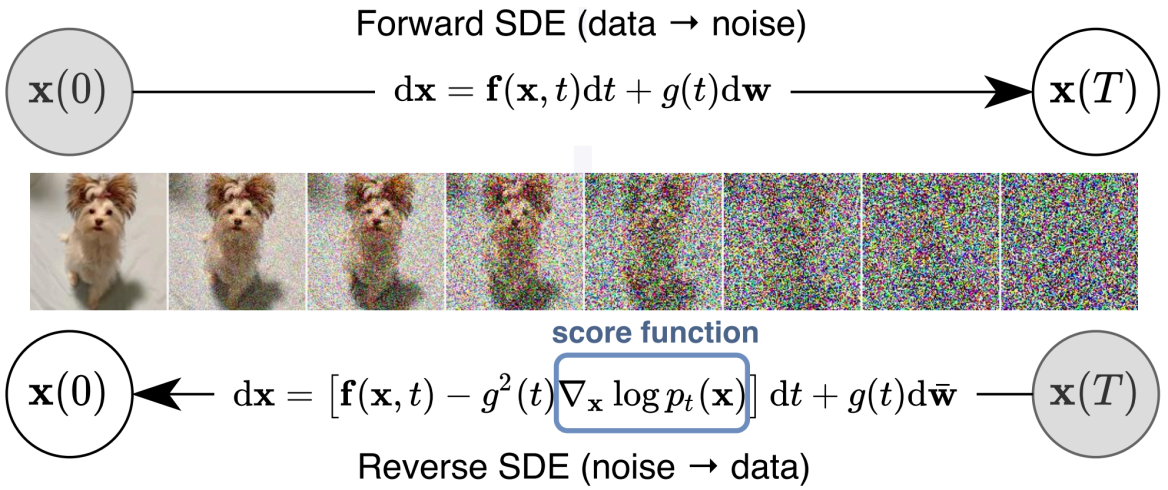
$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0) e^{-\frac{1}{2}\int_0^t \beta(s) ds}, \left(1 - e^{-\int_0^t \beta(s) ds}\right) \cdot \mathbf{I}\right)$$

Sampling with the Reverse SDE

Use the trained score for sampling. Assume terminal prior $\mathcal{N}(0, \mathbf{I})$.

Sampling

1. Sample $\mathbf{x}(1) \sim \mathcal{N}(0, \mathbf{I})$.
2. Solve the reverse SDE using numerical solvers (`SDESolve`).



- Discretizing the reverse SDE provides ancestral sampling.
- Discretizing the probability flow ODE yields deterministic sampling.

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From Time to Noise Level

Use the familiar additive-noise form: $\mathbf{x} = \mathbf{x}_0 + \sigma\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.

$$\mathbf{f}(\mathbf{x}, t) = 0, \qquad g^2(t) = \frac{d\sigma^2(t)}{dt} = 2\sigma(t)\frac{d\sigma(t)}{dt}.$$

Rewrite the probability flow ODE using σ

$$\begin{aligned}\frac{d\mathbf{x}(t)}{dt} &= \mathbf{f}(\mathbf{x}(t), t) - \frac{1}{2}g^2(t)\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t), \sigma(t)) \\ &= -\sigma(t)\frac{d\sigma(t)}{dt}\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t), \sigma(t)).\end{aligned}$$

$$\frac{d\mathbf{x}}{d\sigma} = \left(\frac{d\sigma(t)}{dt}\right)^{-1}\frac{d\mathbf{x}(t)}{dt} = -\sigma\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma).$$

Choose a denoiser using Tweedie's formula

$$\mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}] = \mathbf{x} + \sigma^2\nabla_{\mathbf{x}}\log p_\sigma(\mathbf{x}).$$

$$D_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) \approx \mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}], \qquad \frac{d\mathbf{x}}{d\sigma} = -\sigma\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) = \frac{\mathbf{x} - D_{\boldsymbol{\theta}}(\mathbf{x}, \sigma)}{\sigma}.$$

Decreasing σ moves towards the denoiser prediction. **How do we train this denoiser?**

Train the Denoiser

Use the existing score-training objective. For a noisy-clean pair $(\mathbf{x}, \mathbf{x}_0)$ with $q(\mathbf{x} \mid \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0, \sigma^2 \mathbf{I})$:

$$\mathcal{L} = \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) - \nabla_{\mathbf{x}} \log q(\mathbf{x} \mid \mathbf{x}_0) \right\|_2^2 .$$

Compute the conditional Gaussian target

$$\nabla_{\mathbf{x}} \log q(\mathbf{x} \mid \mathbf{x}_0) = -\frac{\mathbf{x} - \mathbf{x}_0}{\sigma^2} .$$

Substitute the denoiser representation

$$\begin{aligned} \mathcal{L} &= \left\| \frac{D_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) - \mathbf{x}}{\sigma^2} + \frac{\mathbf{x} - \mathbf{x}_0}{\sigma^2} \right\|_2^2 \\ &= \frac{1}{\sigma^4} \left\| D_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) - \mathbf{x}_0 \right\|_2^2 . \end{aligned}$$

Same training objective, different prediction target.
The trained denoiser now supplies the field for sampling.

Generate Samples with a Denoiser

Use the trained denoiser in the ODE solver. Start from strong noise and decrease σ.

Sampling (Euler)

1. Choose $\sigma_0 > \sigma_1 > \cdots > \sigma_N = 0$ and sample $\mathbf{x}^{(0)} \sim \mathcal{N}(0, \sigma_0^2 \mathbf{I})$.
2. For $i = 0, \ldots, N-1$, evaluate $D_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}, \sigma_i)$ and update:

$$\mathbf{x}^{(i+1)} = \mathbf{x}^{(i)} + (\sigma_{i+1} - \sigma_i) \frac{\mathbf{x}^{(i)} - D_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}, \sigma_i)}{\sigma_i}.$$

3. Return $\mathbf{x}^{(N)}$.

We now have an explicit Euler sampler in noise coordinates.
How does this relate to the deterministic DDIM update?

DDIM Is Euler in Noise Coordinates

Recall deterministic DDIM ($s < t$):

$$\hat{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \qquad \hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\hat{\boldsymbol{\epsilon}}}{\sqrt{\bar{\alpha}_t}},$$

$$\mathbf{x}_s = \sqrt{\bar{\alpha}_s}\,\hat{\mathbf{x}}_0 + \sqrt{1-\bar{\alpha}_s}\,\hat{\boldsymbol{\epsilon}}.$$

Recall the DDPM / NCSN rescaling ($\sigma_s < \sigma_t$)

$$\tilde{\mathbf{x}}_u = \frac{\mathbf{x}_u}{\sqrt{\bar{\alpha}_u}}, \qquad \sigma_u = \sqrt{\frac{1-\bar{\alpha}_u}{\bar{\alpha}_u}}, \qquad u \in \{s, t\}.$$

$$D_{\boldsymbol{\theta}}(\tilde{\mathbf{x}}_t, \sigma_t) = \hat{\mathbf{x}}_0 = \tilde{\mathbf{x}}_t - \sigma_t \hat{\boldsymbol{\epsilon}}.$$

Rewrite DDIM as the denoiser Euler step

$$\begin{aligned}\tilde{\mathbf{x}}_s &= \hat{\mathbf{x}}_0 + \sigma_s \hat{\boldsymbol{\epsilon}} = \tilde{\mathbf{x}}_t + (\sigma_s - \sigma_t)\hat{\boldsymbol{\epsilon}} \\ &= \tilde{\mathbf{x}}_t + (\sigma_s - \sigma_t)\frac{\tilde{\mathbf{x}}_t - D_{\boldsymbol{\theta}}(\tilde{\mathbf{x}}_t, \sigma_t)}{\sigma_t}.\end{aligned}$$

DDIM implements this Euler step using the existing noise predictor.

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Continuous-Time Normalizing Flows

Now learn the velocity field itself. Recall $\mathbf{x}_t = \mathbf{x}(t)$ for $t \in [0, 1]$:

- $\mathbf{x}_0 \sim p_0(\mathbf{x})$, where $p_0(\mathbf{x})$ is a base distribution (mostly $\mathcal{N}(0, \mathbf{I})$);
- $\mathbf{x}_1 \sim p_1(\mathbf{x})$, where $p_1(\mathbf{x})$ is the true data distribution $p_{\text{data}}(\mathbf{x})$.

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, t), \quad \text{with initial condition } \mathbf{x}(0) = \mathbf{x}_0.$$

The velocity $\mathbf{v}$ determines a flow $\boldsymbol{\psi}_t$: $\mathbf{x}_t = \boldsymbol{\psi}_t(\mathbf{x}_0)$.

KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x})\right) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

- Likelihood-based CNF training requires ODE solves and Jacobian trace evaluations.
- The adjoint method computes gradients through the ODE solution; repeated solves can be costly.

Continuous-Time Normalizing Flows

KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x})\right) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

- Knowing the vector field $\mathbf{v}(\mathbf{x}, t)$, the KFP (or continuity) equation allows us to compute the density $p_t(\mathbf{x})$.
- Flow matching fits velocities by regression, without model ODE solves or trace evaluations during training.

Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

- Approximate the true vector field $\mathbf{v}(\mathbf{x}, t)$ using $\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)$.
- Use the learned flow $\boldsymbol{\psi}_t$ for deterministic sampling: $\mathbf{x}_1 = \boldsymbol{\psi}_1(\mathbf{x}_0)$.

Flow Matching: Training and Sampling

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})}\left\|\mathbf{v}(\mathbf{x},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

$$\mathbf{x}_0\sim p_0(\mathbf{x})=p(\mathbf{x}),\quad \mathbf{x}_1\sim p_1(\mathbf{x})=p_{\text{data}}(\mathbf{x})$$

Training

1. Sample $t\sim U[0,1]$, $\mathbf{x}_t\sim p_t(\mathbf{x})$.
2. Compute loss $\mathcal{L}=\left\|\mathbf{v}(\mathbf{x}_t,t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t,t)\right\|^2$.

Sampling

1. Sample $\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1=\boldsymbol{\psi}_1(\mathbf{x}_0)=\texttt{ODESolve}_v(\mathbf{x}_0,\boldsymbol{\theta},t_0=0,t_1=1).$$

Problem: The true vector field $\mathbf{v}(\mathbf{x},t)$ is **unknown**.

Flow Matching

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})}\left\|\mathbf{v}(\mathbf{x},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

- There are infinitely many possible $\mathbf{v}(\mathbf{x},t)$ between $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$.
- We need to select the "best" $\mathbf{v}(\mathbf{x},t)$ and make the objective tractable.

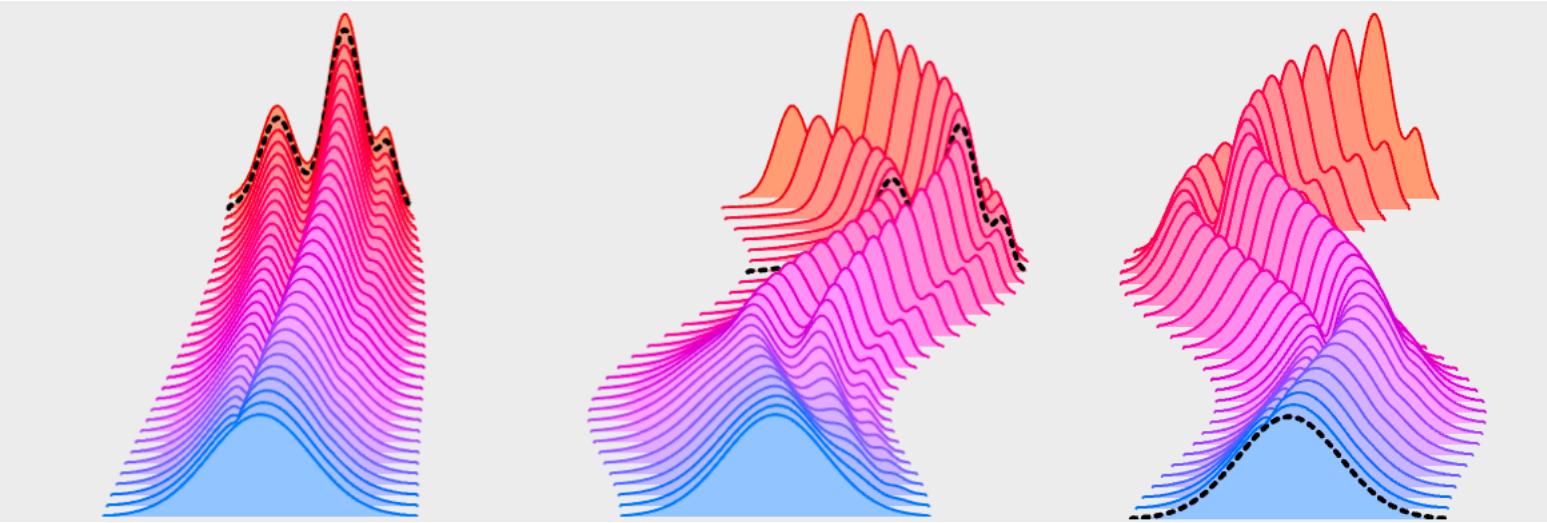


image credit: A Visual Dive into Conditional Flow Matching

Summary

- Score matching (NCSN) and diffusion models (DDPM) are discretizations of SDEs (variance exploding and variance preserving).
- Every SDE admits a corresponding probability flow ODE that follows the same probability path $p_t(\mathbf{x})$, yielding deterministic invertible sampling and log-likelihood evaluation via the instantaneous change of variables.
- A denoiser parameterizes the PF-ODE; DDIM is Euler in rescaled state and noise coordinates. Compare solver cost in network evaluations (NFE).
- SDEs can be reversed in time using the score function $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$; discretizing the reverse SDE gives ancestral sampling, while discretizing the PF-ODE gives deterministic sampling.
- The continuous-time score matching objective generalizes DDPM and NCSN; for affine-drift SDEs the transition kernel $q(\mathbf{x}(t)|\mathbf{x}(0))$ is Gaussian, making the training objective tractable.
- Flow matching trains the velocity field by regression, avoiding model ODE solves and trace evaluations during training.